Chapter 2: Pythagoras Theorem

Q1. Find hypotenuse of a right triangle with sides 9 cm and 12 cm.
Given: Legs = 9 cm and 12 cm
To find: Hypotenuse c
By Pythagoras theorem: $c^2 = 9^2 + 12^2 = 81 + 144 = 225$
$\Rightarrow c = \sqrt{225} = 15$ cm
Answer: Hypotenuse = 15 cm.

Q2. Check whether triangle with sides 7 cm, 24 cm and 25 cm is right-angled.
Given: Sides = 7 cm, 24 cm, 25 cm (largest = 25 cm)
To check: Whether $25^2 = 7^2 + 24^2$
$25^2 = 625$; $7^2 + 24^2 = 49 + 576 = 625$
Since $25^2 = 7^2 + 24^2$, by converse of Pythagoras theorem, the triangle is right-angled.
Answer: Yes, it is a right-angled triangle (right angle opposite side 25 cm).

Q3. Find side of a square whose diagonal is $10\sqrt{2}$ cm.
Given: Diagonal $d = 10\sqrt{2}$ cm
To find: Side a
In a square, diagonal $d = a\sqrt{2}$
$\Rightarrow a = d/\sqrt{2} = (10\sqrt{2})/\sqrt{2} = 10$ cm
Answer: Side of the square = 10 cm.

Q4. Find diagonal of rectangle with length 13 cm and breadth 5 cm.
Given: Length l = 13 cm, breadth b = 5 cm
To find: Diagonal d
Diagonal forms right triangle: $d^2 = l^2 + b^2$
$d^2 = 13^2 + 5^2 = 169 + 25 = 194$
$\Rightarrow d = \sqrt{194}$ cm ≈ 13.93 cm
Answer: Diagonal = $\sqrt{194}$ cm (≈ 13.93 cm).

Q5. A ladder 10 m long reaches a wall at height 8 m. Find distance of its foot from wall.
Given: Ladder (hypotenuse) = 10 m, height reached = 8 m
To find: Distance from wall = x
Right triangle: $10^2 = 8^2 + x^2$
$100 = 64 + x^2$
$x^2 = 36 \Rightarrow x = 6$ m
Answer: Distance of foot from wall = 6 m.

Q6. Find missing side of right triangle if hypotenuse = 26 cm and one side = 10 cm.
Given: Hypotenuse c = 26 cm, one leg a = 10 cm
To find: Other leg b
$c^2 = a^2 + b^2 \Rightarrow 26^2 = 10^2 + b^2$
$676 = 100 + b^2$
$b^2 = 576 \Rightarrow b = 24$ cm
Answer: Missing side = 24 cm.

Q7. Prove Pythagoras theorem.
Statement: In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.
Given: ΔABC right-angled at B. Let AB = a, BC = b, AC = c (hypotenuse).
To prove: $c^2 = a^2 + b^2$
Proof (using similarity):
1) Draw BD ⊥ AC with D on AC.
2) Then $\Delta ABD \sim \Delta ABC$ ($\angle A$ common, both have right angle). Hence: $AB/AC = AD/AB \Rightarrow AB^2 = AC \cdot AD$ ...(1)
3) Also $\Delta CBD \sim \Delta ABC$ ($\angle C$ common, both have right angle). Hence: $BC/AC = DC/BC \Rightarrow BC^2 = AC \cdot DC$ ...(2)
4) Add (1) and (2): $AB^2 + BC^2 = AC(AD + DC)$
5) But AD + DC = AC. Therefore: $AB^2 + BC^2 = AC \cdot AC = AC^2$
$\Rightarrow AB^2 + BC^2 = AC^2$
Thus, $c^2 = a^2 + b^2$. Hence proved.

Q8. Prove converse of Pythagoras theorem.
Statement: If in a triangle, the square of the longest side equals the sum of squares of the other two sides, then the triangle is right-angled.
Given: In ΔABC, let AC be the greatest side and suppose $AC^2 = AB^2 + BC^2$.
To prove: $\angle B = 90°$
Proof:
1) Construct a right-angled triangle ΔABD with $\angle B = 90°$ such that AB is common and BD = BC.
2) Then, by Pythagoras theorem in ΔABD: $AD^2 = AB^2 + BD^2 = AB^2 + BC^2$.
3) Given $AC^2 = AB^2 + BC^2$, so $AC^2 = AD^2 \Rightarrow AC = AD$ (positive lengths).
4) Thus point C and D are on the circle with centre A and radius AC, and also BC = BD.
5) With AB common, BC = BD, AC = AD $\Rightarrow \Delta ABC \cong \Delta ABD$ (SSS).

6) Therefore $\angle ABC = \angle ABD = 90°$.
Hence ΔABC is right-angled at B. Proved.

Q9. Find height of an equilateral triangle of side 12 cm.
Given: Equilateral triangle side = 12 cm
To find: Height h
In an equilateral triangle, altitude bisects the base: base halves = 6 cm each.
Right triangle formed has hypotenuse 12 cm and one side 6 cm.
By Pythagoras: $h^2 = 12^2 - 6^2 = 144 - 36 = 108$
$h = \sqrt{108} = \sqrt{(36\times3)} = 6\sqrt{3}$ cm
Answer: Height = $6\sqrt{3}$ cm.

Q10. Find side of rhombus with diagonals 24 cm and 10 cm.
Given: Diagonals of rhombus = 24 cm and 10 cm
To find: Side s
Property: Diagonals of a rhombus bisect each other at right angles.
So half diagonals: 12 cm and 5 cm form legs of a right triangle, and side is hypotenuse.
$s^2 = 12^2 + 5^2 = 144 + 25 = 169$
$s = \sqrt{169} = 13$ cm
Answer: Side of rhombus = 13 cm.